Dominant Community State Types Reveal Disease-Associated Dominance Patterns in the Human Gut Microbiome

Pawel Gajer[1*†] and Jacques Ravel[1]

[1]University of Maryland School of Medicine

Abstract

Gut microbiome community typing is often built around clustering, making the resulting labels sensitive to modeling choices and harder to interpret at the level of a single sample. We study a deterministic alternative based on L_∞ normalization and dominant-community state types (DCSTs), in which each sample is assigned by rank structure rather than by cluster membership. Using the American Gut Project stool cohort, we analyzed 34,711 samples available in the SILVA species-level table and retained 30,290 after quality control, yielding 274 taxa for downstream analysis. At depth 1, the cohort contained 40 common DCSTs plus a rare-dominant tail; at depth 2, subtype structure revealed additional heterogeneity within broad dominant groups. Among 452 adjusted depth-1 association tests, 28 remained significant after Benjamini-Hochberg correction, and 23 remained significant after a contamination-aware sensitivity analysis. Depth-2 modeling contributed 23 additional adjusted-significant associations, while a comparison against GG2 showed 91.6% exact dominant-genus agreement across overlapping samples. Disease-wise analysis indicated that the clearest signals concentrate in inflammatory and gastrointestinal phenotypes, especially IBD-related structure. External validation in two complementary cohorts, HMP2 and PRJEB84421, supported structural and directional generalization rather than exact AGP-style replication. Together, these results position DCSTs as a practical, interpretable, and hierarchical framework for large-scale gut microbiome analysis.

1. Introduction

Microbiome community typing is useful when it compresses a high-dimensional abundance profile into a smaller number of biologically interpretable states. In practice, however, many typing schemes are built around clustering. That can work well, but it also ties the final labels to choices about transformations, distance metrics, initialization, model class, and the number of clusters. Those choices are not merely technical. They affect whether a reader can understand what a given sample label means and whether two analyses can be compared without a hidden dependence on tuning decisions.

This paper studies a simpler route. A dominant-community state type (DCST) is defined from the rank structure of a sample after L_∞ normalization. At depth 1, the dominant taxon determines the top-level state. At depth 2, the second-ranked taxon refines that state into an ordered subtype.

*Research reported in this publication was supported by grant number INV-048956 from the Gates Foundation.

†Corresponding author: pgajer@gmail.com

The resulting construction is deterministic, hierarchical, and local to the sample: the label is read directly from the ordered dominant coordinates rather than inferred from a fitted global partition.

The gut is a particularly good stress test for this idea. In the vaginal microbiome, strong dominance can be common and even canonical. In the gut, strong dominance is less universal and often appears in dysbiotic or otherwise unusual settings. That makes the gut harder to type cleanly, but also more interesting scientifically. Literature on IBS, IBD, autoimmune disease, and cardiometabolic conditions suggests that shifts in dominant taxa or dominant taxon pairs can mark altered ecological states (Tap et al. 2017; Pittayanon et al. 2019; Lloyd-Price et al. 2019; Gevers et al. 2014; Scher et al. 2013; Vatanen et al. 2018; Z. Wang et al. 2011; Tang et al. 2013).

Our aim is not to argue that dominant taxa alone explain disease biology. The claim is narrower: a clustering-free dominance framework can recover structured gut community states at scale, support disease-wise interpretation without collapsing every phenotype into one pooled signal, and remain informative under covariate adjustment, sensitivity analysis, taxonomy change, and external validation.

What is a DCST?

A DCST is the ordered dominant part of a sample after L_∞ normalization. At depth 1, only the largest component matters. At depth 2, the second-largest component refines the parent class. Samples that do not meet the prevalence threshold for a named dominant type are grouped into `RARE_DOMINANT`.

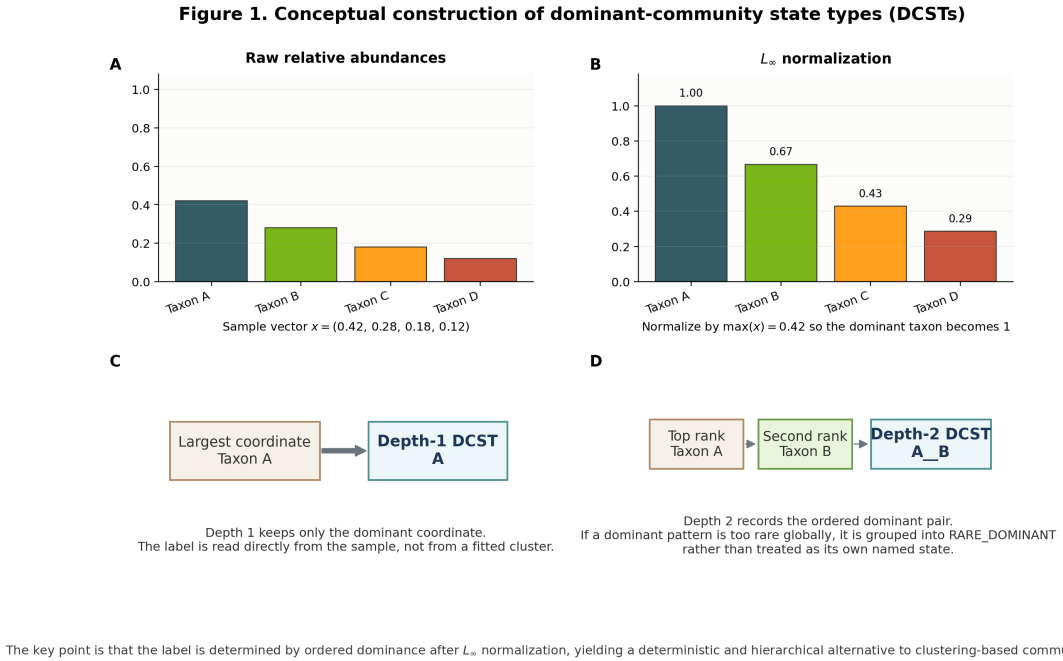

Figure 1: Conceptual schematic of the DCST construction. A sample is first normalized by its largest component, then assigned a depth-1 state by the top-ranked taxon and a depth-2 state by the ordered dominant pair. Rare dominant patterns below the prevalence threshold are grouped into `RARE_DOMINANT`.

2. Methods

2.1 Cohort and preprocessing

The discovery analysis used AGP gut samples from the PRIME-derived SILVA species-level abundance table (McDonald et al. 2018; Quast et al. 2013). The starting table contained 34,711 samples, of which 30,290 passed quality-control filters and were retained for analysis. The retained matrix contained 274 taxa. Age was available for 27,372 samples (90.4%), sex for 25,598 (84.5%), and BMI for 26,541 (87.6%), producing 22,632 complete cases for adjusted modeling.

2.2 DCST construction

Each sample was normalized by its largest component, so the dominant taxon took value 1 and the remaining coordinates were interpreted relative to that reference. A depth-1 DCST was defined by the dominant taxon. A depth-2 DCST was defined by the ordered pair consisting of the dominant and second-ranked taxa. To keep the resulting state space interpretable, only dominant patterns meeting a minimum prevalence threshold n_0 were promoted to named DCSTs; the rest were grouped into `RARE_DOMINANT`.

2.3 Association analysis

We first performed univariate screening with Fisher's exact test. We then fit adjusted logistic models for disease indicators using the DCST label together with available covariates. For a mathematically oriented reader, the odds ratio may be read as follows: an odds ratio above 1 means the DCST is enriched in the given phenotype relative to the reference group, while an odds ratio below 1 means depletion. Multiple testing was handled by the Benjamini-Hochberg procedure (Benjamini and Hochberg 1995), and q-values below 0.05 were treated as statistically notable rather than as proof of mechanism.

2.4 Sensitivity analysis and taxonomy comparison

Because some nominal gut signals can be driven by oral- or skin-associated contaminants, we repeated the adjusted analysis after excluding samples dominated by a prespecified contaminant set. We also ran a parallel dominant-type analysis under GG2 taxonomy (McDonald et al. 2024) and compared dominant-genus assignments between SILVA and GG2 on the overlapping sample set.

2.5 External validation

We used two complementary external cohorts. HMP2 / IBDMDB (`PRJNA398089`) provided clinically grounded inflammatory bowel disease phenotypes but, in the currently available local metadata, was dominated by mucosal gut sites and included repeated measures. PRJEB84421 provided a stool-based inflammatory complement, but was smaller, pediatric, and included orofacial granulomatosis alongside Crohn's disease and healthy controls. The same DCST pipeline was applied to both cohorts with cohort-appropriate prevalence thresholds.

3. Cohort Overview

3.1 Shared AGP overview

The Phase 1 discovery analysis used 30,290 quality-controlled AGP gut samples. At depth 1 with $n_0 = 50$, the cohort yielded 40 named DCSTs plus `RARE_DOMINANT` (1,162 samples). The largest types were `Bacteroides` (10,358 samples), `Escherichia-Shigella` (6,857), `Prevotella_9` (1,870), `Enterobacteriaceae` (1,564), and `Faecalibacterium` (1,320). In other words, the gut landscape is neither one universal dominant state nor an uninterpretable spray of rare labels.

Across the full screen, 59 of 492 univariate Fisher tests were significant after correction. The covariate-adjusted depth-1 pass yielded 28 significant associations across 452 tests, and 23 of those remained significant in the contamination-aware sensitivity analysis. Depth 2 added 23 adjusted-significant subtype associations across 1,340 tests. This common overview is useful because the disease-wise sections below should be read against this shared statistical background rather than as isolated anecdotes. The twelve most common depth-1 states already cover 87.6% of the retained cohort, so the dominant-state picture is concentrated rather than diffusely fragmented (Table 1).

Table 1. Top depth-1 DCSTs in the full AGP cohort.

Rank	DCST	n	Share (%)	Cumulative (%)
1	Bacteroides	10358	34.2	34.2
2	Escherichia-Shigella	6857	22.6	56.8
3	Prevotella_9	1870	6.2	63.0
4	Enterobacteriaceae	1564	5.2	68.2
5	Faecalibacterium	1320	4.4	72.5
6	Rare dominant	1162	3.8	76.4
7	Streptococcus	827	2.7	79.1
8	Pseudomonas	664	2.2	81.3
9	Staphylococcus	583	1.9	83.2
10	Prevotella_7	476	1.6	84.8
11	Unassigned	444	1.5	86.2
12	Lachnospiraceae	416	1.4	87.6

Table 2 collects the single strongest adjusted depth-1 signal for each phenotype with at least one surviving adjusted association. This is not meant to replace the full association screen; it is meant to keep the shared overview legible before the disease-wise sections unpack the same results in context.

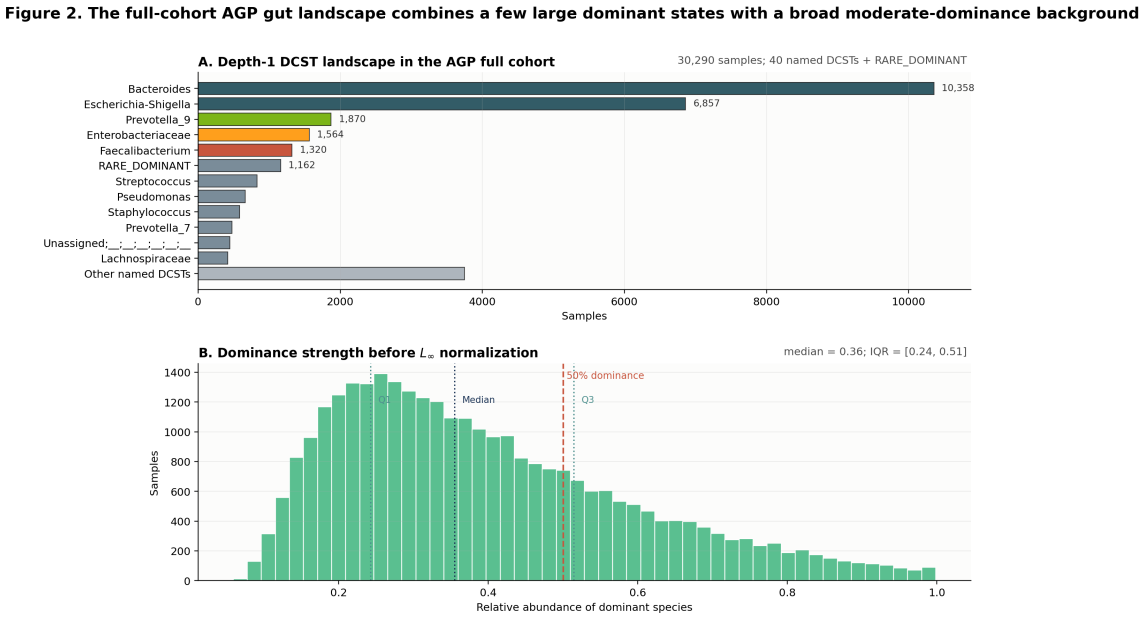

Figure 2: Full-cohort AGP landscape. Panel A shows the leading depth-1 DCSTs after truncation at n0 = 50, while Panel B shows the empirical dominance strength before L_∞ normalization.

Table 2: Headline adjusted depth-1 associations by phenotype. Here C indicates whether the same association remains significant with $q < 0.05$ in the contamination-aware clean analysis.

Phenotype	Direction	DCST	n	Adjusted OR (95% CI)	adj q	C
IBS	Enriched	Escherichia-Shigella	6857	1.23 (1.12-1.35)	6.8e-04	Yes
IBD	Enriched	Morganella	53	16.94 (8.88-32.31)	4.0e-15	Yes
Autoimmune	Enriched	Bacteroides	10358	1.39 (1.28-1.52)	2.1e-11	Yes
Acid reflux	Enriched	Bacteroides	10358	1.28 (1.17-1.40)	6.9e-06	Yes
Seasonal allergies	Depleted	Prevotella_9	1870	0.67 (0.59-0.76)	3.3e-08	Yes
CDI	Enriched	Escherichia-Shigella	6857	1.48 (1.19-1.84)	0.0112	Yes
Lung disease	Enriched	Staphylococcus	583	1.55 (1.17-2.05)	0.0386	No
Migraine	Enriched	Bacteroides	10358	1.16 (1.05-1.26)	0.034	No
Kidney disease	Enriched	Lactobacillus	117	5.85 (2.26-15.14)	0.00779	Yes

Looking only at the counts, the sensitivity analysis reduces the number of adjusted-significant depth-1 associations from 28 to 23. A more informative view is to compare which signals are lost, which are retained, and which become clearer once likely oral- or skin-associated dominants are excluded. Twenty adjusted-significant associations are retained, eight are lost, and three become significant only in the clean analysis. The lost signals are concentrated in `Staphylococcus`- and `Streptococcus`-linked contrasts, while the strongest IBD, allergy, and several `Bacteroides` / `Prevotella_9` associations remain in place. Table 3 shows how this transition looks phenotype by phenotype rather than only in aggregate.

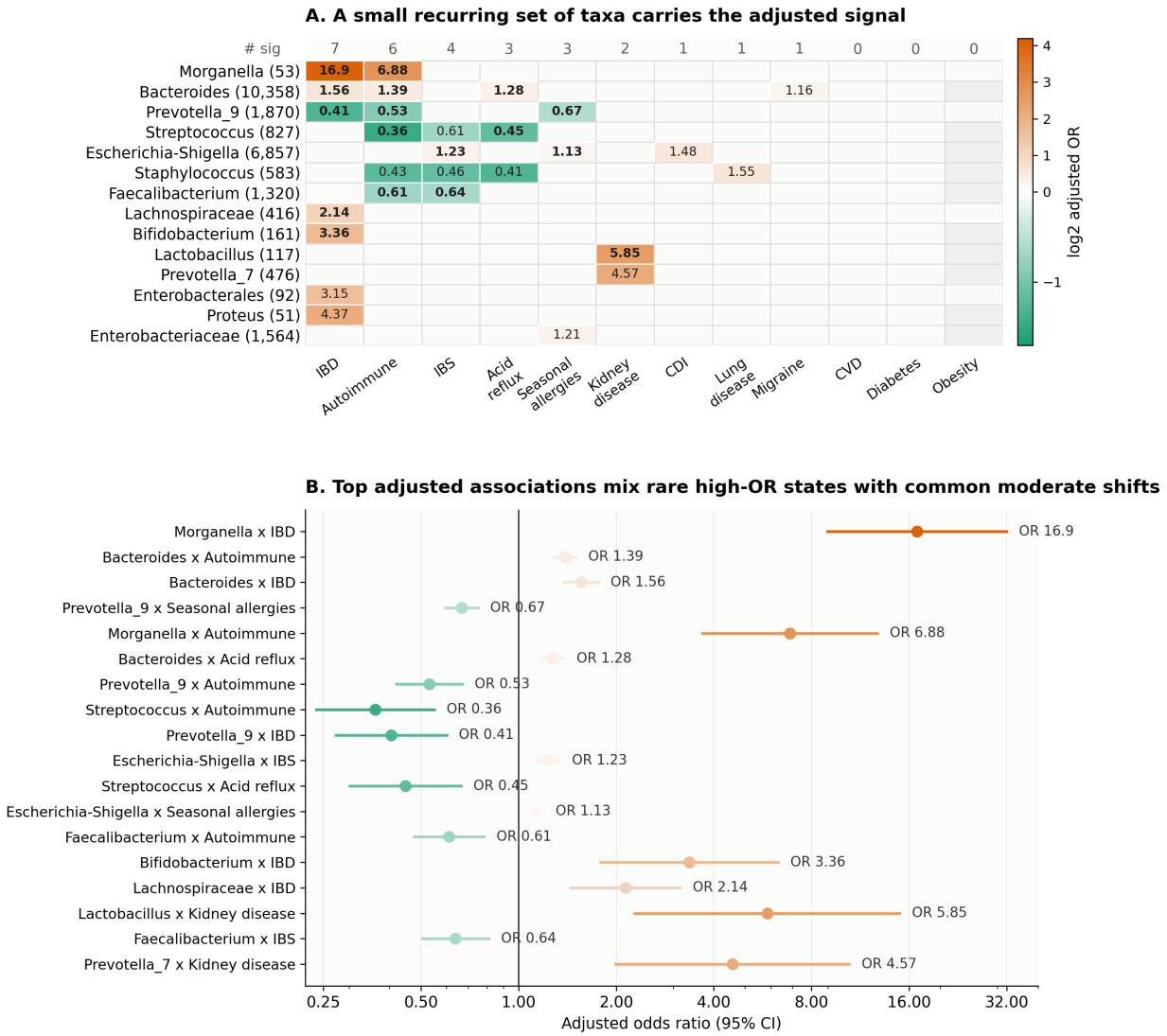

Figure 3: Adjusted depth-1 association overview. Panel A shows all taxa that survive adjustment at least once across the disease screen, with colored cells annotated by adjusted odds ratio and blank cells indicating no surviving adjusted signal. Panel B shows the strongest adjusted associations with 95% confidence intervals.

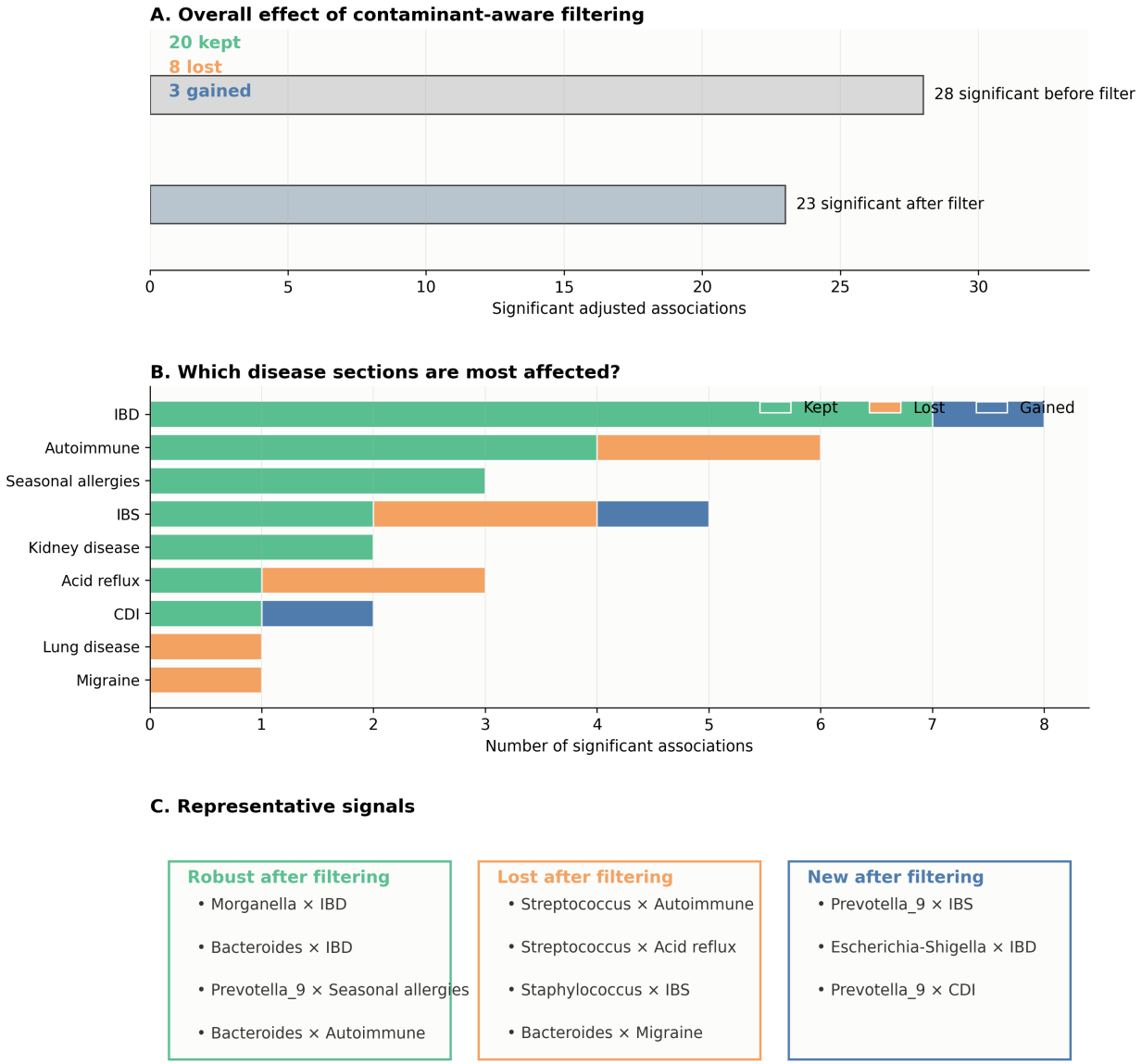

Figure 4: Contamination-aware robustness summary. The clean analysis removes several oral-associated contrasts but preserves the strongest inflammatory and allergy-linked signals.

Table 3: Disease-wise comparison of univariate, adjusted, and clean results. Here n_F, n_A, and n_C denote the numbers of depth-1 associations with Fisher $q < 0.05$, adjusted $q < 0.05$, and clean-analysis $q < 0.05$, respectively.

Phenotype	n_F	n_A	n_C	Representative retained	Representative Fisher-only
IBS	7	4	3	Escherichia-Shigella (1.29 -> 1.23)	Corynebacterium (0.29 -> 0.41)
IBD	9	7	8	Morganella (18.05 -> 16.94)	Prevotella_7 (0.10 -> 0.44)
Autoimmune	9	6	4	Bacteroides (1.34 -> 1.39)	Escherichia-Shigella (1.19 -> 0.99)
Acid reflux	10	3	1	Bacteroides (1.29 -> 1.28)	Prevotella_7 (0.12 -> 0.55)
CVD	6	0	0	–	Pasteurellaceae (8.77 -> 1.74)
Obesity	3	0	0	–	Prevotella_7 (2.75 -> not estimable)
Seasonal allergies	6	3	3	Prevotella_9 (0.75 -> 0.67)	Prevotella_7 (0.22 -> 1.11)
Diabetes	2	0	0	–	Streptococcus (0.28 -> 0.44)
CDI	2	1	2	Escherichia-Shigella (1.45 -> 1.48)	Prevotella_9 (0.47 -> 0.35)
Lung disease	3	1	0	Staphylococcus (1.26 -> 1.55)	Prevotella_7 (0.23 -> 0.89)
Migraine	2	1	0	Bacteroides (1.19 -> 1.16)	Corynebacteriaceae (0.20 -> 0.30)
Kidney disease	0	2	2	Lactobacillus (3.50 -> 5.85)	–

Depth 2 matters most when a broad parent state hides materially different substructures. The depth-2 case-study figure makes that concrete for the depth-1 `Bacteroides` state in IBD. The ten most common depth-2 children cover 85% of all `Bacteroides`-dominant samples, yet their IBD odds ratios range from strong enrichment (`Lachnospiraceae`, `Enterobacteriaceae`) to attenuation or partial reversal (`Faecalibacterium`, `Agathobacter`, `Parabacteroides`). The depth-1 `Bacteroides` signal should therefore be read as an average over heterogeneous subtypes rather than as a single ecological state.

3.2 Taxonomy concordance: SILVA versus GreenGenes2

An independent dominant-type pass under GG2 identified 59 named depth-1 DCSTs plus `RARE_DOMINANT`. On the overlapping 30,290 samples, the exact dominant-genus agreement between SILVA and GG2 was 27,739 / 30,290 (91.6%). This does not mean that the taxonomies are interchangeable. It does mean that the main dominant-typing picture is fairly stable across reference databases, even though some clinically relevant groups, especially `Escherichia-Shigella`, expose genuine naming and resolution differences. The new concordance figure makes an important distinction explicit: harmonized genus-level agreement is high, but exact short-label equality is much lower because GG2 splits some large SILVA states, especially `Bacteroides`, into multiple named groups and leaves `Escherichia-Shigella` largely unresolved. Table 4A summarizes the headline concordance metrics, while Table 4B gives a few representative remappings that matter biologically and interpretively.

Table 4A. Agreement metrics for SILVA versus GG2 depth-1 DCST assignments.

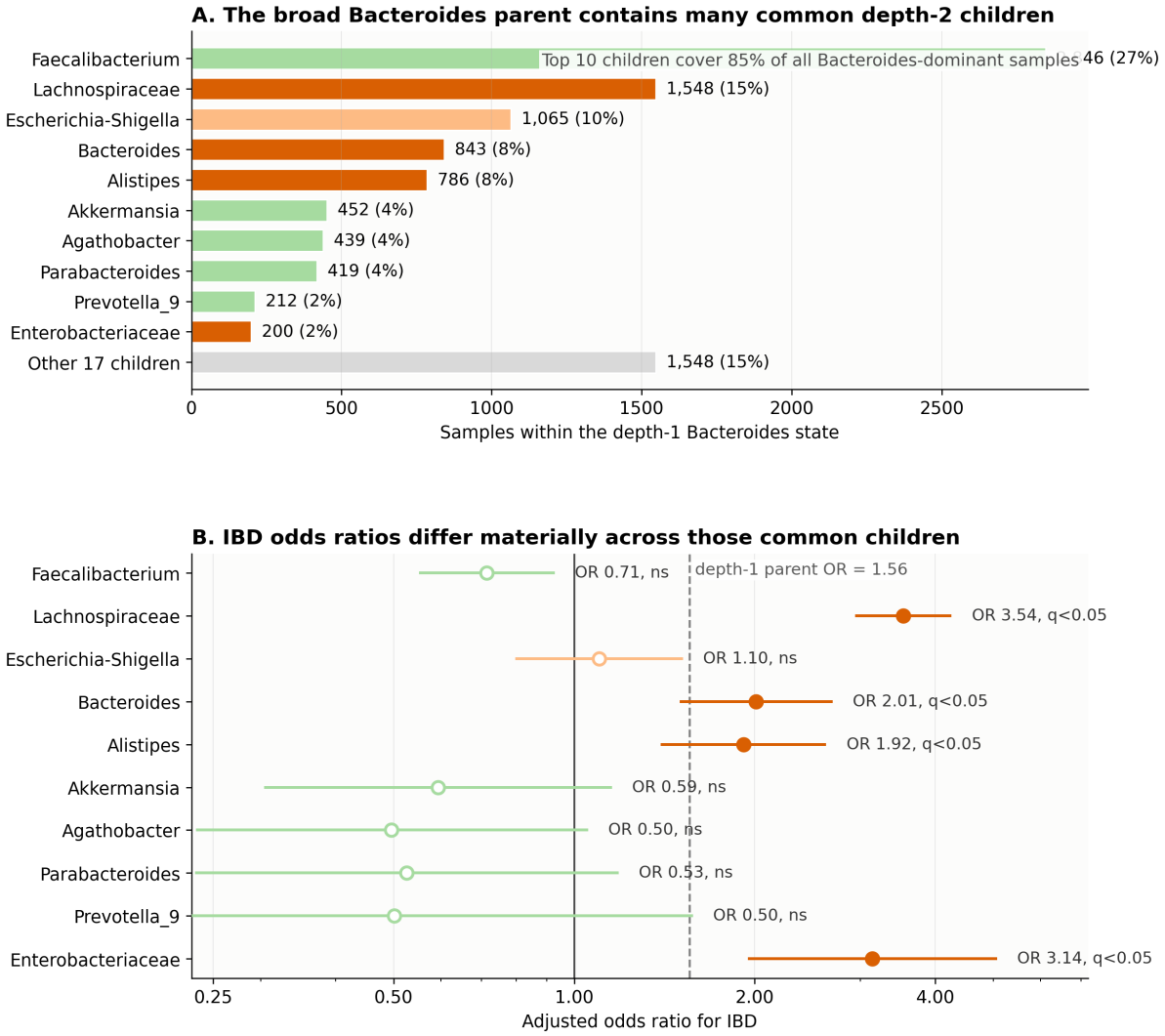

Figure 5: Depth-2 refinement case study for the broad Bacteroides parent in IBD. Panel A shows the most common depth-2 children inside the parent state. Panel B shows that those common children have materially different adjusted odds ratios, with the dashed line marking the coarser depth-1 parent effect.

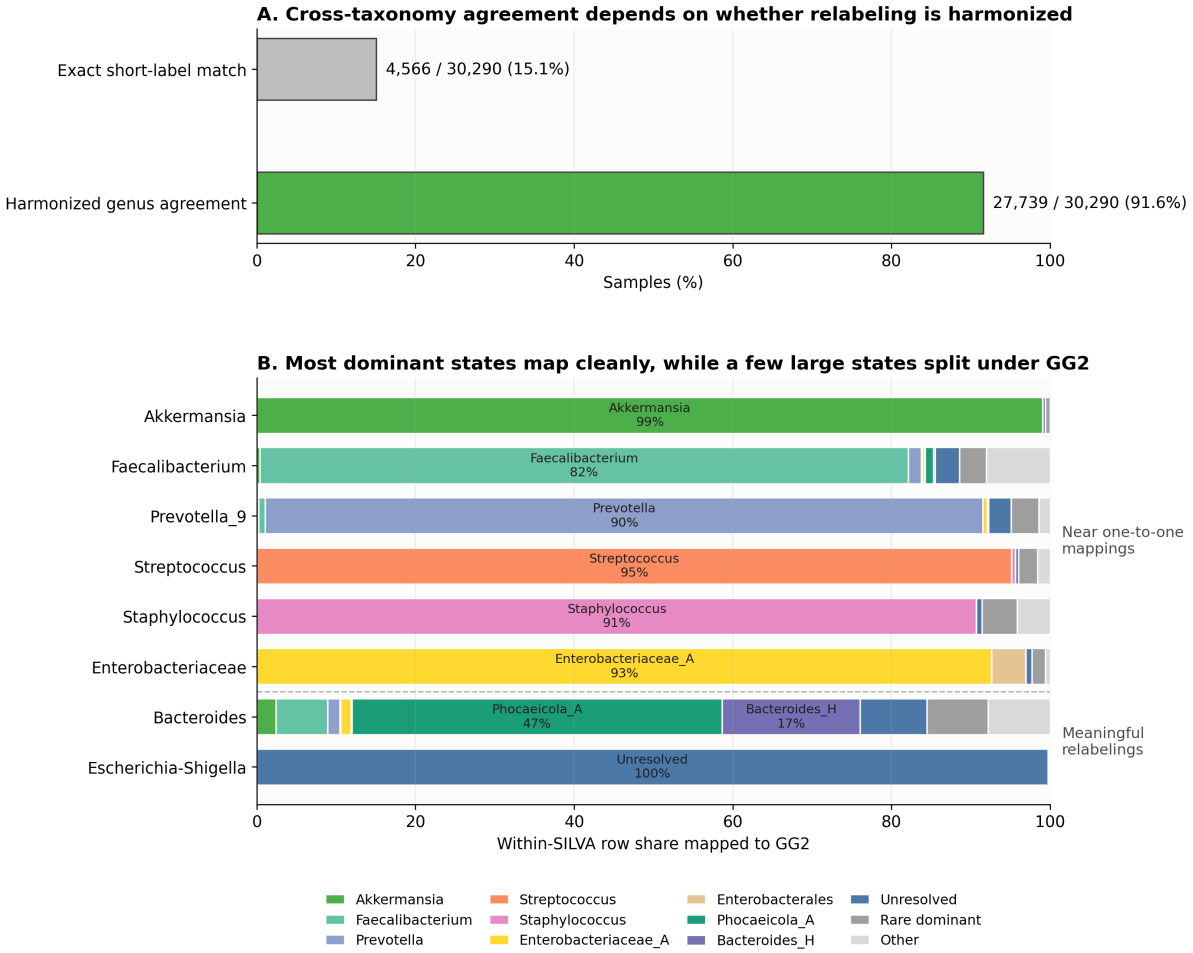

Figure 6: Cross-taxonomy concordance summary. Harmonized agreement between SILVA and GG2 is high, while exact short-label equality is much lower because a few large dominant states are reclassified rather than contradicted.

Metric	Value	Interpretation
Overlapping samples	30,290	Common sample set available for direct SILVA-GG2 comparison.
Harmonized dominant-genus agreement	27,739 / 30,290 (91.6%)	Broad depth-1 landscape is stable after genus-level harmonization.
Exact short-label equality	4,566 / 30,290 (15.1%)	Most disagreements are relabelings or taxonomic splits rather than ecological contradictions.
Named depth-1 states	SILVA: 40 + rare; GG2: 59 + rare	GG2 resolves some large SILVA states into more named groups.

Table 4B. Representative SILVA-to-GG2 remappings.

SILVA state	Main GG2 destination(s)	Row share	Reading
Akkermansia	Akkermansia	99.1%	Near one-to-one mapping.
Faecalibacterium	Faecalibacterium	81.7%	Mostly preserved with modest spillover.
Prevotella_9	Prevotella	90.5%	High genus-level concordance despite label change.
Bacteroides	Phocaeicola_A; Bacteroides_H_857956; unresolved	46.6%; 17.4%; 8.4%	Large SILVA state is split across multiple GG2 labels.
Escherichia-Shigella	Unresolved	99.7%	GG2 leaves this dominant state largely unresolved.
Enterobacteriaceae	Enterobacteriaceae_A_725029; Enterobac-terales_737866	95.6%; 4.3%	Mostly preserved at family/order level with minor split.

4. Disease-Wise Results and Interpretation

The subsections in this section follow a common pattern. Each begins with a brief literature frame, then summarizes the AGP signal across SILVA, GG2, depth 2, and sensitivity analysis, and finally interprets the result at a level that matches the strength of the evidence. This repeated structure is intentional: it lets the reader compare diseases without losing track of which claims are strong, which are tentative, and which are mainly useful for context. Several AGP phenotypes, especially autoimmune disease, seasonal allergies, and lung disease, are best understood as discovery-screen labels rather than harmonized clinical diagnoses, so their discussion leans more on ecological interpretation than on disease-specific inference.

4.1 Irritable bowel syndrome (IBS)

Introduction

IBS is a functional gastrointestinal disorder in which the gut microbiota has been implicated through immune signaling, short-chain fatty acid production, and the microbiota-gut-brain axis (Pittayanon et al. 2019; Dinan and Cryan 2015; Cryan and Dinan 2012). Because many IBS signatures are subtle rather than catastrophically dysbiotic, it is a useful test of whether dominant-state labels retain signal in a noisy population cohort.

Results

In SILVA, the strongest adjusted depth-1 IBS association was enrichment of `Escherichia-Shigella` (adjusted OR 1.23, q = 6.8e-04), accompanied by protective associations for `Faecalibacterium` (adjusted OR 0.64), `Staphylococcus` (adjusted OR 0.46), and `Streptococcus` (adjusted OR 0.61). GG2 showed a qualitatively similar picture, with an enriched unresolved enterobacterial signal and depleted `Staphylococcus` and `Streptococcus` states. At depth 2, additional IBS-related subtype signal appeared in `Escherichia-Shigella`- and `Bacteroides`-centered subtypes. Sensitivity analysis retained `Faecalibacterium`, `Escherichia-Shigella`, and `Prevotella_9`, suggesting that the main IBS pattern is not driven entirely by putative oral contaminants.

Discussion

These IBS findings are consistent with literature linking Enterobacteriaceae expansion and loss of anti-inflammatory commensals to symptom burden (Pittayanon et al. 2019; Tap et al. 2017). The `Faecalibacterium` signal is particularly interpretable for a mechanistic reader, because it points toward loss of butyrate-associated ecological stability rather than toward one IBS-specific pathogen. The oral-associated taxa should still be treated cautiously, and the overall effect sizes are modest compared with IBD.

4.2 Inflammatory bowel disease (IBD)

Introduction

IBD provides the most clinically grounded inflammatory phenotype in the AGP screen. Landmark studies report reduced diversity, expansion of Proteobacteria, and depletion of butyrate-associated commensals, especially *Faecalibacterium prausnitzii* (Frank et al. 2007; Gevers et al. 2014; Lloyd-Price et al. 2019; Sokol et al. 2008; Manichanh et al. 2006).

Results

IBD produced the strongest adjusted depth-1 signal in the entire Phase 1 scan: `Morganella` was highly enriched (adjusted OR 16.94, q = 4.0e-15). Additional adjusted SILVA signals included enrichment of `Bacteroides`, `Lachnospiraceae`, `Bifidobacterium`, `Enterobacterales`, and `Proteus`, along with depletion of `Prevotella_9`. GG2 showed concordant inflammatory structure, again highlighting `Morganella` and related enterobacterial signals. Depth 2 was even more striking: a `Bacteroides` subtype with a `Lachnospiraceae` second-ranked component reached adjusted OR 3.54 with q = 1.6e-37, and multiple IBD-associated subtypes appeared within the broad `Bacteroides` parent class. Eight IBD associations survived sensitivity analysis.

Discussion

IBD is the clearest case here where depth 2 matters both scientifically and mathematically. The established IBD literature emphasizes depletion of obligate anaerobes and butyrate-associated commensals, especially *Faecalibacterium prausnitzii*, together with expansion of inflammatory facultative taxa (Frank et al. 2007; Gevers et al. 2014; Lloyd-Price et al. 2019; Sokol et al. 2008; Sartor 2008). The present DCST results fit that picture in a compressed rank-based language: a rare `Morganella` state marks severe enterobacterial dysbiosis, whereas the broad `Bacteroides` parent resolves into subtypes with materially different odds ratios. That matters for interpretation because the enriched depth-1 `Bacteroides` label is not the opposite of the classic `Faecalibacterium`-depletion story; rather, it partly reflects loss of protective second-rank partners inside Bacteroides-dominant samples. Section 5 strengthens this reading by showing directional depletion of a `Bacteroides__Faecalibacterium` subtype in the external IBD cohorts.

4.3 Autoimmune diseases

Introduction

The AGP autoimmune phenotype is an umbrella label rather than a single disease, so it should be interpreted more cautiously than IBD. Even so, autoimmune disease is an important category because the literature repeatedly links loss of immune-regulatory commensals and expansion of pathobionts to altered Treg/Th17 balance (Scher et al. 2013; Atarashi et al. 2013; De Luca and Shoenfeld 2019; Vatanen et al. 2018).

Results

In SILVA, autoimmune disease showed adjusted enrichment of `Bacteroides` (adjusted OR 1.39, q = 2.1e-11) and `Morganella` (adjusted OR 6.88, q = 1.9e-07), together with depletion of `Prevotella_9`, `Streptococcus`, `Faecalibacterium`, and `Staphylococcus`. GG2 recovered the same broad pattern: enriched inflammatory or enterobacterial structure and depleted `Prevotella` and `Streptococcus` states. Depth 2 again sharpened the signal, especially within `Bacteroides`-centered subtypes, where a major subtype reached adjusted OR 2.25 with q = 1.2e-23. Four autoimmune associations survived sensitivity analysis.

Discussion

Because this AGP label pools multiple autoimmune conditions, the safest reading is an inflammatory umbrella rather than a disease-specific microbiome signature. The observed `Bacteroides` / `Morganella` enrichment and `Faecalibacterium` / `Prevotella_9` depletion are broadly compatible with disease-specific literature in RA and T1D, but the component mechanisms differ: `Prevotella` expansion in RA is not the same biological claim as early-life dysbiosis in T1D, even though both can be placed within a broader Treg/Th17 or immune-education framework (Scher et al. 2013; Vatanen et al. 2018; Atarashi et al. 2013; De Luca and Shoenfeld 2019). The parallel with IBD is therefore informative at the level of shared inflammatory ecology, not at the level of one causal autoimmune taxon.

4.4 Acid reflux / GERD

Introduction

GERD is increasingly discussed in microbiome terms rather than purely as an acid-mediated disorder, with gram-negative-enriched states and loss of gram-positive dominance implicated in esophageal disease (Deshpande et al. 2018; D'Souza et al. 2021; K. Wang et al. 2024).

Results

In SILVA, acid reflux showed adjusted enrichment of `Bacteroides` (adjusted OR 1.28, q = 6.9e-06) and depletion of `Streptococcus` (adjusted OR 0.45, q = 4.2e-03) and `Staphylococcus` (adjusted OR 0.41, q = 1.1e-02). GG2 replicated the depletion of `Streptococcus` and `Staphylococcus` together with a depleted `Prevotella` signal. Depth 2 produced additional `Parabacteroides`- and `Alistipes`-linked signals. After contaminant-aware filtering, only the `Bacteroides` association remained.

Discussion

The reflux results are best interpreted as gut-side correlates of a literature that is itself centered on esophageal and oral compartments (Deshpande et al. 2018; D'Souza et al. 2021). The surviving `Bacteroides` signal therefore suggests a modest reflux-spectrum association rather than a direct esophageal mechanism, whereas the oral-associated taxa are more exposed to compartment choice,

PPI effects, and ecological confounding (K. Wang et al. 2024). The clean analysis is useful here because it narrows the claim to one stool-side association rather than leaving a mixed oral-plus-gut story on the table.

4.5 Cardiovascular disease

Introduction

CVD is one of the most mechanistically developed microbiome literatures because of the TMAO pathway, oral-gut cross-talk, and inflammation-thrombosis links (Z. Wang et al. 2011; Tang et al. 2013; Jie et al. 2017; Zhu et al. 2016).

Results

The univariate CVD signals were strong, especially `Pasteurellaceae`, `Staphylococcus`, and `Neisseria`, but none survived covariate adjustment at depth 1. GG2 showed analogous univariate enrichment of oral-associated or aerotolerant taxa. Depth 2 nonetheless yielded a notable `Agathobacter` association (adjusted OR 3.47, q = 2.6e-05). No CVD signal survived the sensitivity analysis.

Discussion

CVD illustrates why the covariate-adjusted pass matters. Without adjustment, one could easily overstate the oral-associated signals. After adjustment and contaminant-aware filtering, the evidence is much weaker, so this section is best read as hypothesis-generating rather than as a primary biological claim.

4.6 Obesity

Introduction

Obesity has a large and often contradictory microbiome literature, ranging from energy-harvest hypotheses to gene-richness models and `Akkermansia`-linked barrier effects (Turnbaugh et al. 2006; Ley et al. 2006; Le Chatelier et al. 2013; Everard et al. 2013; Walters, Xu, and Knight 2014).

Results

For obesity, the evidence remained at the univariate depth-1 level, led by `Prevotella_7` enrichment (OR 2.75) and weaker shifts in `Escherichia-Shigella` and `Lactobacillus`. No depth-1 association survived covariate adjustment, GG2 mainly reproduced a broad `Prevotella`-side signal, no depth-2 associations reached significance, and nothing survived the sensitivity analysis.

Discussion

These results suggest that dominant-state typing captures some obesity-related structure, but that the signal is either confounded by demographic variables or too diffuse to stabilize at the level of

top-taxon dominance. That is itself useful: it shows where the DCST framework should remain modest rather than forcing every phenotype into a clean association story.

4.7 Seasonal allergies

Introduction

Seasonal allergy is another umbrella phenotype, but one with unusually clear microbiome priors through the hygiene hypothesis and early-life microbiota maturation literature (Strachan 1989; Hua et al. 2016; Hoskinson et al. 2023).

Results

In SILVA, seasonal allergies showed adjusted depletion of `Prevotella_9` (adjusted OR 0.67, q = 3.3e-08) and adjusted enrichment of `Escherichia-Shigella` (adjusted OR 1.13, q = 5.1e-03) and `Enterobacteriaceae` (adjusted OR 1.21, q = 2.7e-02). GG2 showed the same broad pattern, especially strong `Prevotella` depletion. Depth 2 added `Bacteroides`-centered subtype signal, and all three adjusted depth-1 associations survived sensitivity analysis.

Discussion

Among the non-IBD outcomes, seasonal allergy is one of the cleaner screen-level signals in the AGP scan. The protective `Prevotella` pattern fits well with the broader allergy literature, which links allergic phenotypes to lower richness, delayed microbiota maturation, and weaker SCFA-associated immune education (Hua et al. 2016; Hoskinson et al. 2023; Strachan 1989). The Proteobacteria-side enrichment is likewise compatible with a more dysbiotic, less tolerogenic gut ecosystem. Because the signal survives sensitivity analysis and replicates qualitatively in GG2, it is one of the stronger non-IBD findings. Even so, it is best read as an ecological bridge between adult AGP allergy phenotypes and the early-life maturation story, not as a definitive allergic-rhinitis-specific signature.

4.8 Diabetes

Introduction

Diabetes is mechanistically rich but clinically heterogeneous. The literature links diabetes to reduced butyrate producers, endotoxemia, medication effects, and altered fermentation ecology (Qin et al. 2012; Karlsson et al. 2013; Cani et al. 2007; Forslund et al. 2015; Takeuchi et al. 2023).

Results

At depth 1, diabetes yielded only univariate depletion of `Streptococcus` and `Staphylococcus`, with no adjusted-significant depth-1 signal. GG2 again recovered only modest, taxonomically heterogeneous univariate shifts. Depth 2, however, produced a strong `Akkermansia`-associated subtype signal (adjusted OR 4.88, q = 2.1e-04). No diabetes association survived the sensitivity analysis.

Discussion

The `Akkermansia` depth-2 result is biologically interesting precisely because it is paradoxical if interpreted naively. Since metformin can enrich `Akkermansia`, the cross-sectional signal may encode medication rather than disease mechanism. The broader diabetes literature is dominated by T2D, glucose-tolerance gradients, and treatment confounding, so a clean disease-only signal is not expected in a cross-sectional screen of this kind (Qin et al. 2012; Karlsson et al. 2013; Forslund et al. 2015). The present pattern therefore reads sensibly: no stable adjusted depth-1 signal, but one striking `Akkermansia`-associated depth-2 lead that likely mixes disease status, treatment, and carbohydrate-metabolic ecology (Forslund et al. 2015; Takeuchi et al. 2023; Cani et al. 2007).

4.9 *Clostridioides difficile* infection (CDI)

Introduction

CDI is a dysbiosis-driven disease in which colonization resistance is lost after antibiotic disruption, making it a natural target for community-state analysis (Schubert et al. 2014; Seekatz et al. 2014; Sehgal and Khanna 2021).

Results

In SILVA, CDI showed adjusted enrichment of `Escherichia-Shigella` (adjusted OR 1.48, q = 1.1e-02). GG2 highlighted related inflammatory or facultative-anaerobe structure, including `Proteus`. Depth 2 added a strong `Sutterella`-linked subtype signal (adjusted OR 9.85, q = 1.2e-05). After sensitivity analysis, `Escherichia-Shigella` remained enriched and `Prevotella_9` showed a protective pattern.

Discussion

CDI is one of the stronger non-umbrella disease sections because the disease itself is defined by collapse and recovery of colonization resistance (Schubert et al. 2014; Seekatz et al. 2014; Sehgal and Khanna 2021). The enterobacterial enrichment is in line with recurrent-CDI dysbiosis, and the protective `Prevotella_9` direction is at least ecologically compatible with restoration of anaerobic fermentation and bile-acid-converting community structure after FMT. The `Sutterella` depth-2 signal is interesting, but it is best treated as exploratory until it can be tied more directly to recurrence, bile acids, or treatment status.

4.10 Lung disease

Introduction

The gut-lung axis links microbial fermentation, immune tone, and airway phenotypes, but the AGP "lung disease" label is broad and probably pools several different conditions (Budden et al. 2017; Trompette et al. 2014; Horn et al. 2022).

Results

In SILVA, the main adjusted depth-1 lung-disease signal was `Staphylococcus` enrichment (adjusted OR 1.55, q = 3.9e-02), while univariate depletion appeared for `Prevotella_7`, `Streptococcus`, and `Pasteurellaceae`. GG2 reproduced the depleted `Prevotella` / `Streptococcus` / `Pasteurellaceae` side of the signal. Depth 2 added `Agathobacter` and `Staphylococcus` subtype associations. None of the signals survived sensitivity analysis.

Discussion

This phenotype is too broad to support a strong disease-specific claim, and the comparison literature itself is split between gut-origin asthma studies, airway COPD studies, and respiratory-infection models (Budden et al. 2017; Trompette et al. 2014; Horn et al. 2022). The surviving `Staphylococcus` signal is therefore difficult to interpret as a gut-lung-axis result per se, especially because it disappears in the clean analysis. The most defensible reading is that the AGP screen detects a weak, partly compartment-mixed respiratory signal rather than a coherent lung-disease microbiome signature.

4.11 Migraine

Introduction

Migraine is increasingly discussed through gut-brain signaling, inflammation, and microbial metabolite effects, but the current microbiome evidence base is thinner than for IBD or allergy (Cryan and Dinan 2012; Gorenshtein et al. 2025; Kappéter et al. 2023).

Results

Migraine produced one adjusted depth-1 signal in SILVA: `Bacteroides` enrichment (adjusted OR 1.16, q = 3.4e-02). GG2 showed a qualitatively similar `Bacteroides`-side enrichment together with weaker secondary signals. No depth-2 association reached significance, and nothing survived the sensitivity analysis.

Discussion

This remains exploratory. The direct migraine microbiome literature is still small and heterogeneous, with current reviews emphasizing dysbiosis, SCFA and barrier pathways, and gut-brain-axis signaling more than any one reproducible taxon (Gorenshtein et al. 2025; Kappéter et al. 2023; Cryan and Dinan 2012). The present `Bacteroides` enrichment is therefore best read as a weak cross-disease inflammatory echo rather than as migraine-specific evidence.

4.12 Kidney disease

Introduction

The gut-kidney axis is mechanistically attractive because microbial metabolites such as TMAO, indoxyl sulfate, and p-cresyl sulfate can worsen renal decline (Tang et al. 2015; Vaziri et al. 2013).

Results

Kidney disease showed no univariate depth-1 signals, but the adjusted pass identified strong enrichment of `Lactobacillus` (adjusted OR 5.85, q = 7.8e-03) and `Prevotella_7` (adjusted OR 4.57, q = 1.0e-02). GG2 recovered a compatible pattern through enriched `Prevotella` and a reclassified `Bacteroides` group. Depth 2 retained a `Lactobacillus` signal, and both adjusted depth-1 associations survived sensitivity analysis.

Discussion

Kidney disease offers a useful example of a signal that is statistically stronger than its immediate biological label. The CKD literature emphasizes TMAO, uremic toxins, and depletion of saccharolytic commensals more than it emphasizes `Lactobacillus` as a universal pathogenic signal (Tang et al. 2015; Vaziri et al. 2013). The observed `Lactobacillus` and `Prevotella_7` enrichments may therefore encode treatment behavior, probiotic use, or compensatory ecology superimposed on a true gut-kidney-axis signal. What keeps this section prominent is robustness: unlike several weaker phenotypes, the adjusted depth-1 associations survive the clean analysis.

5. External Validation Across Two Complementary Cohorts

5.1 HMP2 / IBDMDB

HMP2 / IBDMDB (`PRJNA398089`) contributed the clinically grounded inflammatory validation arm. After phenotype filtering, the local cohort contained 166 samples: 45 healthy controls, 81 Crohn's disease samples, and 40 ulcerative colitis samples. The caveat is ecological: in the local metadata, the retained set is dominated by rectal, ileal, and colonic sites and includes repeated measures. At depth 1, the HMP2 landscape was too sparse to yield strong multiple-testing-corrected disease signal. At depth 2, however, a `Bacteroides__Faecalibacterium` subtype was depleted in `IBD_vs_Healthy` (OR = 0.391, q = 0.016), `UC_vs_Healthy` (OR = 0.318, q = 0.032), and `Crohn_vs_Healthy` (OR = 0.429, q = 0.041).

5.2 PRJEB84421

PRJEB84421 supplied the stool-based complement. It retained 73 samples split across 20 healthy controls, 24 Crohn's disease samples, and 29 pediatric orofacial granulomatosis samples. This cohort is a better ecological match to AGP stool than HMP2, but it is smaller and phenotype-mixed. Its strongest signals were therefore more modest and cohort-specific: `RARE_DOMINANT` was enriched in

`OFG_vs_Healthy` at depth 1 (OR = 4.93, q = 0.062) and in the combined inflammatory-vs-healthy comparison at depth 2 (OR = 4.94, q = 0.084).

Table 5. External validation cohort summary.

Cohort	n	Phenotype split	Main caveat	Main signal	Interpretation
HMP2 / IBDMDB	166	45 Healthy / 81 Crohn / 40 UC	Mucosal gut sites; repeated measures	No depth-1 q<0.05; depth-2 depletion of a Bacteroides/Faecalibacterium subtype	Directional / structural replication
PRJEB84421 / OFGCD-FI-2025	73	20 Healthy / 24 Crohn / 29 OFG	Stool-based but pediatric and phenotype-mixed	Rare-dominant enrichment in OFG vs healthy and inflammatory vs healthy	Stool-based complement with cohort-specific signal

5.3 Interpretation

Taken together, the two external cohorts support a narrower but more defensible validation claim than exact taxon-by-taxon replication. HMP2 shows that the DCST framework captures clinically relevant inflammatory subtype structure in an IBD-focused longitudinal cohort, while PRJEB84421 shows that the same framework remains interpretable in a smaller stool-based inflammatory cohort. The validation therefore supports **structural and directional generalization** rather than exact AGP-style replication. In practice, this strengthens the IBD-centered part of the Phase 1 story much more than it strengthens the more tentative umbrella outcomes.

6. General Discussion

The disease-wise organization is not merely editorial. It prevents the paper's biological interpretation from being flattened into one pooled list of odds ratios. Once the results are viewed disease by disease, a more coherent picture emerges. IBD, seasonal allergies, IBS, CDI, and kidney disease carry the clearest depth-1 or depth-2 signals. Autoimmune disease is also strong statistically, but its AGP phenotype is broad and should be interpreted as an umbrella inflammatory label. CVD, obesity, lung disease, diabetes, and migraine are more mixed: they contain usable signals, but those signals are either attenuated by adjustment, sensitive to contamination-aware filtering, or too coarse for confident mechanistic interpretation.

Several cross-cutting patterns survive this disease-wise reading. First, `Bacteroides`-dominant structure repeatedly aligns with inflammatory or inflammation-adjacent phenotypes, including IBD, autoimmune disease, acid reflux, and migraine. Second, `Prevotella`-side dominance is repeatedly

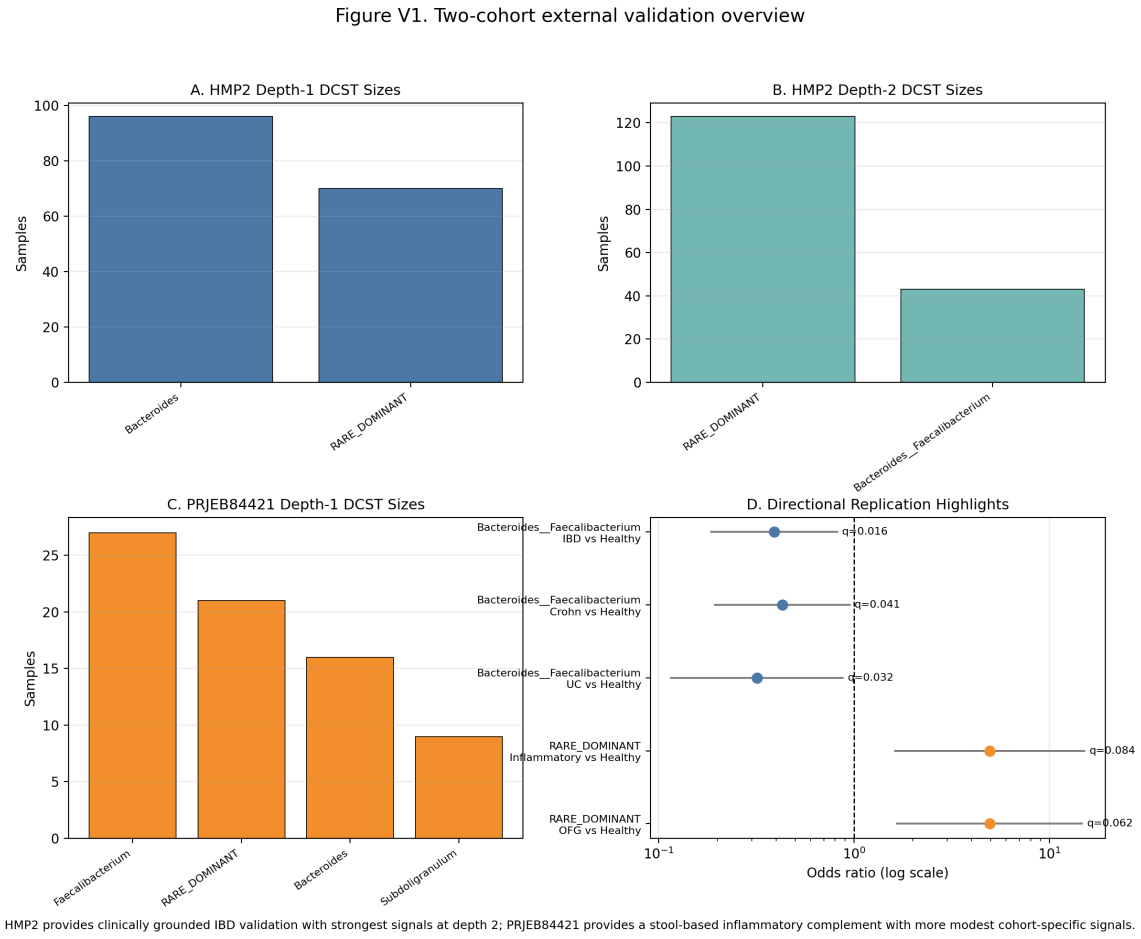

Figure 7: Two-cohort external validation summary. HMP2 provides clinically grounded IBD validation, while PRJEB84421 provides a stool-based inflammatory complement. Together they support structural and directional generalization rather than exact replication.

protective in allergy and inflammatory settings, especially seasonal allergies and IBD. Third, rare but striking enterobacterial states, particularly `Morganella`, behave as markers of severe dysbiosis rather than as subtle population shifts. Fourth, depth 2 adds real information by turning a top-dominant label into an ordered ecological pair. For a mathematically oriented reader, this is important: the refinement is explicit and interpretable rather than latent.

Those broad patterns also line up with the stronger disease-specific literatures in recognizable ways. In IBD, the combination of enterobacterial rare states, `Prevotella_9` depletion, and depth-2 `Bacteroides` heterogeneity matches the treatment-naive and multi-omics literature that emphasizes loss of butyrate- associated anaerobes together with inflammatory expansion of facultative taxa (Gevers et al. 2014; Lloyd-Price et al. 2019; Sokol et al. 2008). IBS shows a weaker but directionally similar ecology, with Enterobacteriaceae-side enrichment and `Faecalibacterium` depletion rather than a dramatic dysbiosis collapse (Pittayanon et al. 2019; Tap et al. 2017). Seasonal allergy aligns with the microbiota-maturation and immune-education literature, in which reduced tolerogenic fermentation and delayed ecological maturation accompany allergic phenotypes (Hua et al. 2016; Hoskinson et al. 2023; Strachan 1989). CDI fits the colonization-resistance framework of bile acids, donor-like recovery, and Proteobacteria blooms (Schubert et al. 2014; Seekatz et al. 2014; Sehgal and Khanna 2021). Kidney disease is the least mechanistically settled of these stronger signals, but it still sits plausibly within the broader gut-kidney axis defined by TMAO and uremic-toxin biology (Tang et al. 2015; Vaziri et al. 2013).

The paper should still be careful about overclaiming. AGP phenotypes are self-reported. Some disease bins are umbrella categories. The analysis is cross-sectional, so reverse causation, medication effects, and residual confounding remain live possibilities. The sensitivity analysis shows that some signals rely on taxa that are plausibly oral or skin associated. And the validation cohorts, while valuable, are intentionally heterogeneous rather than perfect one-to-one replicas of AGP stool. Those limitations argue for a measured conclusion: DCSTs organize the data in a biologically useful way, but they do not by themselves establish mechanism or causality. For the same reason, the disease-specific literature cited in Section 4 should be read as mechanistic context for the discovery-screen phenotypes rather than as proof that the AGP labels are clinically identical to those literature-defined entities.

7. Conclusion

Taken together, this Phase 1 manuscript is best understood as a disease-wise atlas built on a common dominant-state framework. DCSTs recover a structured gut landscape, remain fairly stable across taxonomic schemas, and identify a nontrivial set of phenotype associations that survive adjustment and, in many cases, contamination-aware filtering. The strongest part of the overall story concerns inflammatory phenotypes, especially IBD and related subtype structure, and that part is now supported by two complementary external validation cohorts. This is already enough to justify DCSTs as a useful analytical language for future gut microbiome studies and for the broader `linf` application paper.

8. Data and Code Availability

The manuscript source, manuscript figures, and manuscript table builders used for this draft are maintained in the public `linf` repository: https://github.com/pgajer/linf. The discovery analysis is based on the American Gut Project stool cohort (`PRJEB11419`) through PRIME-derived species-level abundance tables, and the external validation analyses use `PRJNA398089` (HMP2 / IBDMDB) and `PRJEB84421`. The specific processed result tables used to generate the figures and Tables 1-4 are those analyzed in the companion gut-microbiome working tree during manuscript preparation.

9. Funding

Research reported in this publication was supported by grant number `INV-048956` from the Gates Foundation.

10. Competing Interests

The authors declare no competing interests.

11. Acknowledgments

We thank the participants and investigators of the American Gut Project, HMP2 / IBDMDB, and PRJEB84421 for making these cohort resources available, and we thank the PRIME project context that made the large-scale gut screen feasible.

12. References

Atarashi, Koji, Taira Tanoue, Keiji Oshima, et al. 2013. "Treg Induction by a Rationally Selected Mixture of Clostridia Strains from the Human Microbiota." *Nature* 500: 232–36. https://doi.org/10.1038/nature12331.

Benjamini, Yoav, and Yosef Hochberg. 1995. "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing." *Journal of the Royal Statistical Society Series B* 57: 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x.

Budden, Kurtis F, Shaan L Gellatly, David L A Wood, et al. 2017. "Emerging Pathogenic Links Between Microbiota and the Gut–Lung Axis." *Nature Reviews Microbiology* 15: 55–63. https://doi.org/10.1038/nrmicro.2016.142.

Cani, Patrice D, Jacques Amar, Miguel A Iglesias, et al. 2007. "Metabolic Endotoxemia Initiates Obesity and Insulin Resistance." *Diabetes* 56: 1761–72. https://doi.org/10.2337/db06-1491.

Cryan, John F, and Timothy G Dinan. 2012. "Mind-Altering Microorganisms: The Impact of the Gut Microbiota on Brain and Behaviour." *Nature Reviews Neuroscience* 13: 701–12. https://doi.org/10.1038/nrn3346.

D'Souza, S M, K Houston, L Keenan, B S Yoo, P J Parekh, and D A Johnson. 2021. "Role of Microbial Dysbiosis in the Pathogenesis of Esophageal Mucosal Disease: A Paradigm Shift from

Acid to Bacteria?" *World Journal of Gastroenterology* 27: 2054–72. https://doi.org/10.3748/wjg.v27.i18.2054.

De Luca, F, and Y Shoenfeld. 2019. "The Microbiome in Autoimmune Diseases." *Clinical and Experimental Immunology* 195: 74–85. https://doi.org/10.1111/cei.13158.

Deshpande, N P, S M Riordan, N Castaño-Rodríguez, M R Wilkins, and N O Kaakoush. 2018. "Signatures Within the Esophageal Microbiome Are Associated with Host Genetics, Age, and Disease." *Microbiome* 6: 227. https://doi.org/10.1186/s40168-018-0611-4.

Dinan, Timothy G, and John F Cryan. 2015. "Irritable Bowel Syndrome: A Microbiome-Gut-Brain Axis Disorder?" *Neurogastroenterology and Motility* 27: 445–55. https://doi.org/10.1111/nmo.12520.

Everard, Amandine, Clara Belzer, Lucie Geurts, et al. 2013. "Cross-Talk Between *Akkermansia Muciniphila* and Intestinal Epithelium Controls Diet-Induced Obesity." *Proceedings of the National Academy of Sciences* 110: 9066–71. https://doi.org/10.1073/pnas.1219451110.

Forslund, Kristoffer, Falk Hildebrand, Trine Nielsen, et al. 2015. "Disentangling Type 2 Diabetes and Metformin Treatment Signatures in the Human Gut Microbiota." *Nature* 528: 262–66. https://doi.org/10.1038/nature15766.

Frank, Daniel N, April L St Amand, Robert A Feldman, et al. 2007. "Molecular-Phylogenetic Characterization of Microbial Community Imbalances in Human Inflammatory Bowel Diseases." *Proceedings of the National Academy of Sciences* 104: 13780–85. https://doi.org/10.1073/pnas.0706625104.

Gevers, Dirk, Subra Kugathasan, Lee A Denson, et al. 2014. "The Treatment-Naive Microbiome in New-Onset Crohn's Disease." *Cell Host and Microbe* 15: 382–92. https://doi.org/10.1016/j.chom.2014.02.005.

Gorenshtein, Alon, Kamel Shihada, Liron Leibovitch, Tom Liba, and Avner Goren. 2025. "The Association Between Migraine and Gut Microbiota: A Systematic Review." *Acta Neurologica Belgica* 125: 977–87. https://doi.org/10.1007/s13760-025-02779-y.

Horn, Kadi J, Melissa A Schopper, Zoe G Drigot, and Sarah E Clark. 2022. "Airway *Prevotella* Promote TLR2-Dependent Neutrophil Activation and Rapid Clearance of *Streptococcus Pneumoniae* from the Lung." *Nature Communications* 13: 3321. https://doi.org/10.1038/s41467-022-31074-0.

Hoskinson, C L, D L Y Dai, K L Del Bel, et al. 2023. "Delayed Gut Microbiota Maturation in the First Year of Life Is a Hallmark of Pediatric Allergic Disease." *Nature Communications* 14: 4785. https://doi.org/10.1038/s41467-023-40336-4.

Hua, Xinmin, James J Goedert, Angela Pu, Guoqin Yu, and Jianxin Shi. 2016. "Allergy Associations with the Adult Fecal Microbiota: Analysis of the American Gut Project." *EBioMedicine* 3: 172–79. https://doi.org/10.1016/j.ebiom.2015.11.038.

Jie, Zhuye, Huihua Xia, Shi-Long Zhong, et al. 2017. "The Gut Microbiome in Atherosclerotic Cardiovascular Disease." *Nature Communications* 8: 845. https://doi.org/10.1038/s41467-017-00900-1.

Kappéter, Ágnes, Dávid Sipos, Adorján Varga, et al. 2023. "Migraine as a Disease Associated with Dysbiosis and Possible Therapy with Fecal Microbiota Transplantation." *Microorganisms* 11:

2083. https://doi.org/10.3390/microorganisms11082083.

Karlsson, Fredrik H, Valentina Tremaroli, Intawat Nookaew, et al. 2013. "Gut Metagenomics in European Women with Normal, Impaired and Diabetic Glucose Control." *Nature* 498: 99–103. https://doi.org/10.1038/nature12198.

Le Chatelier, Emmanuelle, Trine Nielsen, Junjie Qin, et al. 2013. "Richness of Human Gut Microbiome Correlates with Metabolic Markers." *Nature* 500: 541–46. https://doi.org/10.1038/nature12506.

Ley, Ruth E, Peter J Turnbaugh, Samuel Klein, and Jeffrey I Gordon. 2006. "Human Gut Microbes Associated with Obesity." *Nature* 444: 1022–23. https://doi.org/10.1038/4441022a.

Lloyd-Price, Jason, Cesar Arze, Ashwin N Ananthakrishnan, et al. 2019. "Multi-Omics of the Gut Microbial Ecosystem in Inflammatory Bowel Diseases." *Nature* 569: 655–62. https://doi.org/10.1038/s41586-019-1237-9.

Manichanh, Chaysip, Laurence Rigottier-Gois, Emmanuelle Bonnaud, et al. 2006. "Reduced Diversity of Faecal Microbiota in Crohn's Disease Revealed by a Metagenomic Approach." *Gut* 55: 205–11. https://doi.org/10.1136/gut.2005.073817.

McDonald, Daniel, Embriette Hyde, Justine W Debelius, et al. 2018. "American Gut: An Open Platform for Citizen Science Microbiome Research." *mSystems* 3: e00031–18. https://doi.org/10.1128/mSystems.00031-18.

McDonald, Daniel, Yueyu Jiang, Metin Balaban, et al. 2024. "GreenGenes2 Unifies Microbial Data in a Single Reference Tree." *Nature Biotechnology* 42: 715–18. https://doi.org/10.1038/s41587-023-02009-5.

Pittayanon, Rapat, Jennifer T Lau, Yuhong Yuan, Grigorios I Leontiadis, Frances Tse, Michael G Surette, and Paul Moayyedi. 2019. "Gut Microbiota in Patients with Irritable Bowel Syndrome: A Systematic Review." *Gastroenterology* 157: 97–108. https://doi.org/10.1053/j.gastro.2019.03.049.

Qin, Junjie, Yingrui Li, Zhiming Cai, et al. 2012. "A Metagenome-Wide Association Study of Gut Microbiota in Type 2 Diabetes." *Nature* 490: 55–60. https://doi.org/10.1038/nature11450.

Quast, Christian, Elmar Pruesse, Pelin Yilmaz, et al. 2013. "The SILVA Ribosomal RNA Gene Database Project: Improved Data Processing and Web-Based Tools." *Nucleic Acids Research* 41: D590–96. https://doi.org/10.1093/nar/gks1219.

Sartor, R Balfour. 2008. "Microbial Influences in Inflammatory Bowel Diseases." *Gastroenterology* 134: 577–94. https://doi.org/10.1053/j.gastro.2007.11.059.

Scher, Jose U, Andrew Sczesnak, Randy S Longman, et al. 2013. "Expansion of Intestinal *Prevotella Copri* Correlates with Enhanced Susceptibility to Arthritis." *eLife* 2: e01202. https://doi.org/10.7554/eLife.01202.

Schubert, Allison M, Matthew A M Rogers, Corrie Ring, et al. 2014. "Microbiome Data Distinguish Patients with *Clostridium Difficile* Infection and Non-*C. Difficile*-Associated Diarrhea from Healthy Controls." *mBio* 5: e01021–14. https://doi.org/10.1128/mBio.01021-14.

Seekatz, Anna M, Jorn Aas, Christine E Gessert, et al. 2014. "Recovery of the Gut Microbiome Following Fecal Microbiota Transplantation." *mBio* 5: e00893–14. https://doi.org/10.1128/mBio.00893-14.

Sehgal, Kanika, and Sahil Khanna. 2021. "Gut Microbiome and *Clostridioides Difficile* Infection: A Closer Look at the Microscopic Interface." *Therapeutic Advances in Gastroenterology* 14: 1756284821994736. https://doi.org/10.1177/1756284821994736.

Sokol, Harry, Bernardo Pigneur, Laetitia Watterlot, et al. 2008. "*Faecalibacterium Prausnitzii* Is an Anti-Inflammatory Commensal Bacterium Identified by Gut Microbiota Analysis of Crohn Disease Patients." *Proceedings of the National Academy of Sciences* 105: 16731–36. https://doi.org/10.1073/pnas.0804812105.

Strachan, D P. 1989. "Hay Fever, Hygiene, and Household Size." *BMJ* 299: 1259–60. https://doi.org/10.1136/bmj.299.6710.1259.

Takeuchi, Takeshi, Tetsuya Kubota, Yumiko Nakanishi, et al. 2023. "Gut Microbial Carbohydrate Metabolism Contributes to Insulin Resistance." *Nature* 621: 591–98. https://doi.org/10.1038/s41586-023-06466-x.

Tang, W H Wilson, Zeneng Wang, David J Kennedy, et al. 2015. "Gut Microbiota-Dependent Trimethylamine N-Oxide (TMAO) Pathway Contributes to Both Development of Renal Insufficiency and Mortality Risk in Chronic Kidney Disease." *Circulation Research* 116: 448–55. https://doi.org/10.1161/CIRCRESAHA.116.305360.

Tang, W H Wilson, Zeneng Wang, Bruce S Levison, et al. 2013. "Intestinal Microbial Metabolism of Phosphatidylcholine and Cardiovascular Risk." *New England Journal of Medicine* 368: 1575–84. https://doi.org/10.1056/NEJMoa1109400.

Tap, Julien, Muriel Derrien, Henrik Törnblom, Romain Brazeilles, Sonia Cools-Portier, Joël Doré, Sanna Störsrud, Benjamin Le Nevé, and Magnus Simrén. 2017. "Identification of an Intestinal Microbiota Signature Associated with Severity of Irritable Bowel Syndrome." *Gastroenterology* 152: 111–23. https://doi.org/10.1053/j.gastro.2016.09.049.

Trompette, Aurélien, Evan S Gollwitzer, Khyati Yadava, et al. 2014. "Gut Microbiota Metabolism of Dietary Fiber Influences Allergic Airway Disease and Hematopoiesis." *Nature Medicine* 20: 159–66. https://doi.org/10.1038/nm.3444.

Turnbaugh, Peter J, Ruth E Ley, Michael A Mahowald, Vincent Magrini, Elaine R Mardis, and Jeffrey I Gordon. 2006. "An Obesity-Associated Gut Microbiome with Increased Capacity for Energy Harvest." *Nature* 444: 1027–31. https://doi.org/10.1038/nature05414.

Vatanen, Tommi, Eric A Franzosa, Randall Schwager, et al. 2018. "The Human Gut Microbiome in Early-Onset Type 1 Diabetes from the TEDDY Study." *Nature* 562: 589–94. https://doi.org/10.1038/s41586-018-0620-2.

Vaziri, Nosratola D, Jakk Wong, Madeleine Pahl, et al. 2013. "Chronic Kidney Disease Alters Intestinal Microbial Flora." *Kidney International* 83: 308–15. https://doi.org/10.1038/ki.2012.345.

Walters, William A, Zech Xu, and Rob Knight. 2014. "Meta-Analyses of Human Gut Microbes Associated with Obesity and IBD." *FEBS Letters* 588: 4223–33. https://doi.org/10.1016/j.febslet.2014.09.039.

Wang, K, S Wang, Y Chen, X Lu, D Wang, Y Zhang, W Pan, C Zhou, and D Zou. 2024. "Causal Relationship Between Gut Microbiota and Risk of Gastroesophageal Reflux Disease: A Genetic Correlation and Bidirectional Mendelian Randomization Study." *Frontiers in Immunology* 15:

1327503. https://doi.org/10.3389/fimmu.2024.1327503.

Wang, Zeneng, Erin Klipfell, Brian J Bennett, et al. 2011. "Gut Flora Metabolism of Phosphatidylcholine Promotes Cardiovascular Disease." *Nature* 472: 57–63. https://doi.org/10.1038/nature09922.

Zhu, W, J C Gregory, E Org, et al. 2016. "Gut Microbe-Generated Trimethylamine N-Oxide from Dietary Choline Is Prothrombotic in Subjects." *Cell* 165: 111–24. https://doi.org/10.1016/j.cell.2016.02.011.